

Feasibility Memo: Triage Data Assessment

To: Mercer General ED Board

From: Sekou Ruddock

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Subject: Data Feasibility Assessment for Hybrid Clinical-Text Decision Support System (HCT-DSS)

1. Verdict

The historical dataset is suitable for developing a baseline triage decision-support model, provided we strictly respect the noted limitations regarding external validity and local Caribbean deployment.

2. Dataset Summary

I have processed and profiled a historical emergency department dataset containing over 560,000 distinct patient encounters. For each encounter, I analyzed approximately 225 patient characteristics ranging from basic demographics (age, gender) to specific vital signs (heart rate, oxygen saturation, temperature) to predict the patient's official Emergency Severity Index (ESI) on a scale of 1 (resuscitation) to 5 (non-urgent clinic care). The dataset also includes roughly 200 specific "chief complaint" flags indicating why the patient arrived (e.g., chest pain, trauma, shortness of breath); however, these flags are highly sparse, meaning many patients do not have a standard complaint checked off in the system. The patient mix skews heavily toward mid-acuity (ESI 3) cases, which accurately reflects a standard emergency department caseload.

3. Top 3 Quality Concerns & Mitigations

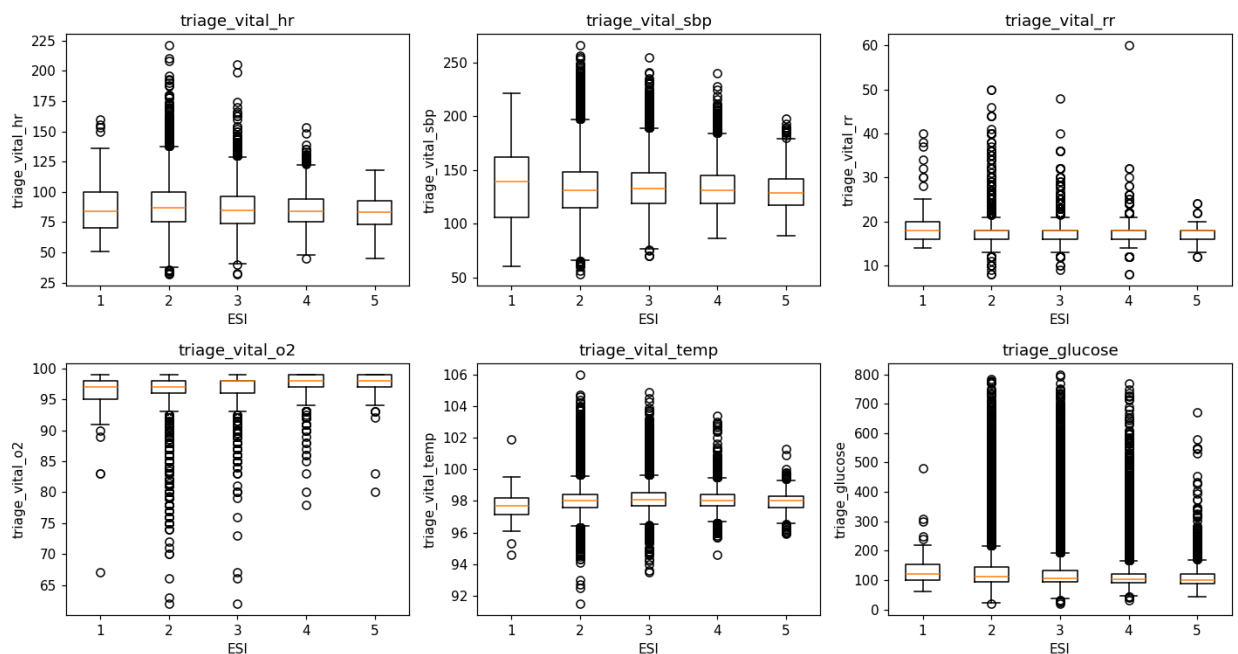
- **Information Available Only After Triage (Future Leakage):** Certain columns in the raw data, such as a patient's final hospital admission status (disposition), contain information that is only known *after* the medical encounter is resolved. If we feed this into our algorithm, it gives the system the "answer key" ahead of time, making it look artificially highly accurate during testing but completely useless during a live, front-door triage shift.
 - Mitigation: I aggressively audited the dataset and permanently excluded all future-looking columns. The model will only be trained on data points that a triage nurse physically possesses at the moment of intake.
- **Gaps in Patient Records (Missingness):** In a chaotic, high-volume emergency environment, not every data field is filled out for every patient. Our data profiling revealed significant gaps in certain historical details and secondary vital signs.
 - Mitigation: I dropped columns that were missing more than 80% of the time, as they hold no reliable predictive value. For the remaining, manageable gaps, we filled the missing fields with typical, neutral baseline values and logged every instance. This ensures the system will not freeze or crash when a nurse inputs an incomplete patient profile during a busy shift.

Feasibility Memo: Triage Data Assessment

- **Mismatch in Patient Populations (Representativeness):** This dataset originates from a massive, highly resourced United States academic hospital. The demographics, underlying population health, and resource availability do not perfectly mirror the walk-in population of a stand-alone Caribbean facility like Mercer General.
 - Mitigation: This data will be treated solely as a foundational training ground. Before any live deployment, the resulting model must be validated against a localized, historical dataset from Mercer General to ensure it safely captures our unique patient mix.

4. Top 3 Reasons to Proceed

- **A Genuine Triage Target:** Unlike many commercial systems that use financial billing costs as a flawed proxy for how sick a patient is, this dataset includes the actual, human-assigned ESI levels. This means we are training the system to replicate real clinical urgency, not administrative spending.
- **Strong, Distinct Physiological Signals:** As demonstrated in our data dashboard, the critical vital signs naturally separate the acuity levels exactly as clinical intuition would expect. For example, the sickest patient cohorts (ESI 1 and 2) show massive, distinct variations in their respiratory rates and oxygen saturation compared to stable walk-ins. The data mathematically supports basic emergency medicine principles.
- **A Fully Documented, Auditable Pipeline:** Every single decision made during the data profiling and cleaning phase has been tracked. Whether we dropped a corrupted column or smoothed out a physically impossible temperature reading, the exact rule and rationale are recorded in the attached cleaning log. This provides a transparent audit trail for the medical board and regulatory reviewers.



Feasibility Memo: Triage Data Assessment

5. Caveats

We must acknowledge several hard limits in this data before proceeding. First, the temperature values are recorded entirely in Fahrenheit; we will need to build an automated conversion pipeline to align this with Mercer General's standard Celsius workflow so that local nurses are not confused by the outputs. Second, we must be cautious regarding the 200 chief-complaint flags. Because they are so sparsely populated, the model may struggle to learn from them reliably, and we cannot assume a missing flag means the patient does not have that symptom. Finally, as noted in our concerns, the external validity for a Caribbean ED remains untested. A model that performs flawlessly on this US dataset must still be treated with extreme clinical suspicion until it proves itself on Mercer General's historical records.

Top 10 Triage Feature Shortlist

The following features represent the most promising data points for our baseline model, ranked by a combination of statistical correlation and clinical emergency relevance.

1. Triage Vital SpO2 (Oxygen Saturation)

- Justification: Hypoxia is a rapid, life-threatening emergency requiring immediate resuscitation, and our correlation matrix proves a massive statistical relationship between low oxygen saturation and the most critical ESI 1 assignments.

2. Triage Vital RR (Respiratory Rate)

- Justification: A spiked respiratory rate is universally taught as the earliest physiological warning sign of systemic deterioration; our data dashboard confirms that abnormal breathing rates clearly and distinctly separate the highest-acuity patients from the rest of the waiting room.

3. Triage Vital HR (Heart Rate)

- Justification: Extreme tachycardia or bradycardia immediately signals severe trauma, shock, or systemic infection to a triage nurse, and the dataset shows a highly reliable variance in heart rates for ESI 1 and 2 patients compared to non-urgent cases.

4. Age

- Justification: Elderly patients possess far less physiological reserve and can decompensate rapidly from seemingly minor complaints; the demographic distributions in our analysis confirm that advanced age heavily skews a patient's probability toward a more urgent triage tier.

5. Triage Vital SBP (Systolic Blood Pressure)

- Justification: A crashing systolic blood pressure is the hallmark of circulatory shock, while extreme highs indicate stroke risk; the data strongly links these extreme SBP anomalies to rapid bed-placement decisions.

Feasibility Memo: Triage Data Assessment

6. CC_ChestPain (Chief Complaint: Chest Pain)

- Justification: Chest pain is a universal "red-flag" protocol symptom for acute coronary syndrome and pulmonary embolisms; our cross-tabulation charts show a dense, statistically significant grouping of this complaint resulting in immediate high-acuity (ESI 2) sorting.

7. CC_ShortnessOfBreath (Chief Complaint: Shortness of Breath)

- Justification: Severe respiratory distress cannot wait in a lobby chair; the dataset confirms a strong predictive correlation between this specific symptom flag and the necessity for immediate clinical intervention.

8. Triage Vital Temp (Temperature)

- Justification: Fever is the primary objective indicator of systemic infection or sepsis; although it requires conversion to Celsius for our local staff, it acts as a reliable mathematical divider between low-risk mechanical injuries and high-risk infectious presentations.

9. Triage Vital DBP (Diastolic Blood Pressure)

- Justification: A triage nurse requires diastolic pressure alongside systolic to fully understand a patient's total organ perfusion state; our correlation analysis demonstrates that it works synergistically with SBP to accurately flag high-risk cardiovascular patients.

10. CC_AbdominalPain (Chief Complaint: Abdominal Pain)

- Justification: While often a vague symptom, it represents a massive volume of ED presentations that frequently hide fatal surgical emergencies (like appendicitis or ruptured aneurysms); the statistical profiling shows this complaint is a heavy, accurate driver for mid-to-high acuity (ESI 2-3) sorting.

Github Link- <https://github.com/Colosolizer/carisurg-portfolio>